Support Vector Machines

Instructions: Show all intermediate steps. Unless otherwise stated, labels are $y_i \in \{-1, +1\}$, the intercept is unregularized, and

$$f(x) = w^\top x + b.$$

For canonical hard-margin SVMs, use

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1.$$

Part A: Geometry and Classification

1. Decision boundary and distances

A classifier uses

$$w = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \qquad b = -10.$$

Consider

$$x_A = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad x_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad x_C = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

(a) Write the equation of the decision boundary.

(b) Calculate the decision score for each point.

(c) Classify each point as positive, negative, or on the boundary.

(d) Calculate each point's perpendicular distance from the boundary.

(e) Find the closest point on the boundary to x_A.

2. Functional and geometric margins

A classifier has

$$w = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad b = 1.$$

The labelled observations are

$$x_1 = (3, 1)^\top, \quad y_1 = +1,$$
$$x_2 = (1, 2)^\top, \quad y_2 = -1,$$
$$x_3 = (0, 0)^\top, \quad y_3 = +1.$$

(a) Calculate the functional margin of every observation.

(b) Calculate the signed geometric margin of every observation.

(c) Determine the minimum functional and geometric margins.

(d) Replace (w, b) by $(5w, 5b)$ and repeat the margin calculations.

(e) Explain mathematically which margins change and which remain unchanged.

3. Canonical margin hyperplanes

A canonical SVM classifier has

$$w = (2, -1)^\top, \qquad b = -3.$$

(a) Write the decision boundary and both margin hyperplanes.

(b) Calculate the distance from the decision boundary to each margin hyperplane.

(c) Calculate the full margin width.

(d) For $x = (3, 1)^\top$ with label $+1$, determine whether the point lies outside the margin, on the margin, or inside the margin.

(e) Repeat for $x = (1, 0)^\top$ with label -1.

4. One-dimensional hard-margin SVM

The training observations are

x_i	-3	-1	2	4
y_i	-1	-1	$+1$	$+1$

(a) Identify the observations closest to the separating boundary.

(b) Use their canonical margin equations to obtain w and b.

(c) Verify all four hard-margin constraints.

(d) Find the decision boundary and full margin width.

(e) Classify the test points $x = 0$ and $x = 1$.

5. Two-dimensional maximum-margin separator

Consider

x_i	y_i
$(2, 0)^\top$	$+1$
$(0, 2)^\top$	$+1$
$(0, 0)^\top$	-1
$(-1, -1)^\top$	-1

(a) Determine the canonical maximum-margin values of w and b.

(b) Identify the observations lying on the margin hyperplanes.

(c) Calculate the full margin width.

(d) Verify the signed functional margins of all observations.

(e) Classify $(1, 1)^\top$ and $(0.2, 0.2)^\top$.

Part B: Soft Margins and Optimization

For this section, use

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

subject to

$$y_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

6. Slack variables and hinge loss

For a fixed classifier, the decision scores and labels are

i	1	2	3	4	5
y_i	$+1$	$+1$	-1	-1	$+1$
$f(x_i)$	1.5	0.4	-0.8	0.3	-0.6

(a) Calculate each signed functional margin.

(b) Calculate the smallest feasible slack for each observation.

(c) Identify correctly classified observations inside the margin.

(d) Identify misclassified observations.

(e) If $\|w\|^2 = 4$ and $C = 2$, calculate the primal objective using the smallest feasible slacks.

7. Comparing candidate classifiers

Three feasible classifiers have the following quantities, using their smallest feasible slack variables:

Classifier	$\|w\|^2$	$\sum_i \xi_i$
A	2	3
B	8	1
D	14	0

(a) Write the objective of each classifier as a function of C.

(b) Find all pairwise equality values of C.

(c) Determine which candidate has the smallest objective for each interval of $C > 0$.

(d) Identify any values of C at which multiple candidates tie.

(e) Explain the trade-off between weight norm and violations.

8. One subgradient update

A linear SVM uses

$$J(w, b) = \frac{1}{2}\|w\|^2 + C \sum_i \max\{0, 1 - y_i(w^\top x_i + b)\}.$$

At the current iteration,

$$w = (1, -1)^\top, \qquad b = 0, \qquad C = 1.$$

The observations are

$$x_1 = (2, 0)^\top, \quad y_1 = +1,$$
$$x_2 = (0, 2)^\top, \quad y_2 = -1,$$
$$x_3 = (0, 0)^\top, \quad y_3 = +1.$$

(a) Calculate the signed margins and identify active hinge losses.

(b) Calculate a subgradient with respect to w and b.

(c) Perform one simultaneous update with learning rate $\eta = 0.1$.

(d) Calculate the updated decision scores.

(e) Evaluate the objective before and after the update.

9. KKT conditions and support vectors

For four observations at a soft-margin optimum with $C = 2$, suppose

i	y_i	$y_i f(x_i)$	α_i
1	+1	1.5	0
2	+1	1.0	1
3	−1	0.4	2
4	+1	−0.2	2

(a) Find the smallest feasible slack for every observation.

(b) Calculate $\mu_i = C - \alpha_i$, where μ_i is the multiplier for $\xi_i \geq 0$.

(c) Check both complementary-slackness conditions:

$$\alpha_i\big[y_i f(x_i) - 1 + \xi_i\big] = 0, \qquad \mu_i \xi_i = 0.$$

(d) Check the intercept stationarity condition $\sum_i \alpha_i y_i = 0$.

(e) Identify the free support vector, bounded support vectors, and non-support vector.

10. Solving a soft-margin SVM through its dual

There are two one-dimensional observations:

$$x_1 = -1, \quad y_1 = -1, \qquad x_2 = 1, \quad y_2 = +1.$$

Use a linear kernel.

(a) Write the dual objective and constraints.

(b) Use the equality constraint to express both multipliers using a single variable α.

(c) Find the optimal α as a function of $C > 0$.

(d) Recover w and determine a valid choice of b.

(e) For $C = \frac{1}{4}$ and $C = 1$, calculate the smallest feasible slacks and primal objective.

(f) For both values of C, calculate the dual objective and compare it with the primal objective.

Part C: Kernels and Nonlinear Classification

11. Polynomial feature mapping

Consider

$$K(x, z) = (x^\top z)^2, \qquad x, z \in \mathbb{R}^2.$$

(a) Construct an explicit feature map $\phi(x)$ satisfying $K(x, z) = \phi(x)^\top \phi(z)$.

(b) For $x = (1, 2)^\top$ and $z = (2, -1)^\top$, calculate $K(x, z)$ directly.

(c) Calculate $\phi(x)$ and $\phi(z)$ and verify their inner product.

(d) A feature-space classifier uses

$$w_\phi = (1, -1, 0)^\top, \qquad b = 0,$$

with feature coordinates ordered as $(x_1^2, x_2^2, \sqrt{2}x_1x_2)$. Write its decision boundary in the original input space.

(e) Classify $(2, 1)^\top$ and $(1, 2)^\top$.

12. XOR separation in feature space

The observations are

$$(1, 1)^\top,\ (-1, -1)^\top \quad \text{with label } +1,$$
$$(1, -1)^\top,\ (-1, 1)^\top \quad \text{with label } -1.$$

Use

$$\phi(x) = x_1 x_2.$$

(a) Show algebraically why no affine linear separator in the original two-dimensional space can satisfy all hard-margin constraints.

(b) Map all four observations into feature space.

(c) Find the canonical hard-margin classifier $f(x) = w_\phi \phi(x) + b$.

(d) Calculate its full margin width in feature space.

(e) Describe the decision boundary in the original input space.

13. RBF kernel prediction

A trained RBF SVM has two support vectors:

$$x_1 = (0,0)^\top, \quad y_1 = -1, \quad \alpha_1 = 1,$$

$$x_2 = (2,0)^\top, \quad y_2 = +1, \quad \alpha_2 = 1.$$

The kernel and intercept are

$$K(x,z) = \exp\left(-\frac{1}{2}\|x - z\|^2\right), \qquad b = 0.$$

(a) Write the decision function.

(b) Calculate the scores for $(0,0)^\top$, $(1,0)^\top$, and $(2,0)^\top$.

(c) Give the corresponding classifications, identifying any point on the boundary.

(d) Derive the complete decision boundary in input space.

(e) Determine the limit of the score for $(t,0)^\top$ as $t \to +\infty$.

14. Testing a proposed kernel matrix

A proposed similarity function produces

$$G = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

on three inputs.

(a) Calculate the eigenvalues of G.

(b) Determine whether G is positive semidefinite.

(c) Find a nonzero vector v for which $v^\top G v < 0$, if such a vector exists.

(d) Explain what this implies about the proposed similarity being a positive-semidefinite kernel on these inputs.

(e) Find the smallest $\lambda \geq 0$ for which $G + \lambda I$ is positive semidefinite.

Part D: Support Vector Regression

15. Epsilon-insensitive regression loss

An SVR model uses

$$f(x) = 2x + 1, \qquad \varepsilon = 0.5.$$

The observations are

x_i	0	1	2	3
y_i	1.2	2.1	5.4	8.0

Use the constraints

$$y_i - f(x_i) \le \varepsilon + \xi_i,$$
$$f(x_i) - y_i \le \varepsilon + \xi_i^*, \qquad \xi_i, \xi_i^* \ge 0.$$

(a) Calculate the prediction and residual $r_i = y_i - f(x_i)$ for each observation.

(b) Calculate each epsilon-insensitive loss

$$L_\varepsilon(r_i) = \max\{0, |r_i| - \varepsilon\}.$$

(c) Calculate the smallest feasible ξ_i and ξ_i^*.

(d) Identify observations strictly inside the tube and those outside it.

(e) For $C = 2$, calculate

$$\frac{1}{2}w^2 + C\sum_i(\xi_i + \xi_i^*).$$

(f) Recalculate the total loss when ε is increased to 1, keeping the prediction function fixed.